

Sequential Task:

Online Store

Object-Oriented Programming and GUI

April 9, 2026

1. Declare the enum type `DiscountType`, which represents different discount levels expressed as percentages.
 - (a) Add a private, immutable field `discountPercent` of type `int`, which stores the discount value for a given constant.
 - (b) Implement the constructor `DiscountType(int discountPercent)`, whose purpose is to assign the discount value to the class field.
 - i. Accept the parameter `discountPercent`.
 - ii. Assign its value to the field `discountPercent`.
 - (c) Add the method `getDiscountPercent()`, which returns the value of the field `discountPercent`.
 - (d) Implement the method `applyDiscount(BigDecimal price)`, whose purpose is to calculate the price after applying the discount.
 - i. Check whether the value of the field `discountPercent` is equal to 0.
 - ii. If the discount is 0, return the unchanged value `price`.
 - iii. Otherwise, calculate the discount multiplier as `1 - discountPercent / 100.0` and store it as a `BigDecimal`.
 - iv. Multiply the value `price` by the calculated multiplier.
 - v. Set the scale of the result to 2 decimal places using the rounding mode `RoundingMode.HALF_UP`.
 - vi. Return the calculated value as the discounted price.
 - (e) Override the method `toString()` so that it returns a readable representation of the discount.
 - If the value of `discountPercent` is 0, return a string indicating that no discount applies.
 - Otherwise, return a string containing the minus sign and the percentage discount value.
 - (f) Define enum constants corresponding to different discount levels, assigning each of them the percentage value passed to the constructor:
 - `NONE` - 0,
 - `SALE_5` - 5,
 - `SALE_10` - 10,

- SALE_15 - 15,
 - CLEARANCE_20 - 20,
 - CLEARANCE_30 - 30,
 - CLEARANCE_50 - 50,
 - LIQUIDATION_70 - 70;
2. Declare the enum type **AvailabilityStatus**, which represents different product availability states.
- (a) Add a private, immutable field **description** of type **String**, which stores the description of the availability state.
 - (b) Implement the constructor **AvailabilityStatus(String description)**, whose purpose is to initialize the description field.
 - i. Accept the parameter **description**.
 - ii. Assign its value to the field **description**.
 - (c) Add the method **getDescription()**, which returns the value of the field **description**.
 - (d) Implement the public static method **determineStatus(int quantity)**, whose purpose is to determine the availability status based on the number of available units.
 - Accept the parameter **quantity**, representing the number of available products.
 - Check whether the value of **quantity** is equal to 0.
 - If so, return **OUT_OF_STOCK**.
 - Otherwise, check whether the value of **quantity** is less than or equal to 5.
 - If so, return **LOW_STOCK**.
 - Otherwise, return **AVAILABLE**.
 - (e) Define enum constants corresponding to availability states, assigning each of them a text description passed to the constructor
 - **AVAILABLE** - Available,
 - **LOW_STOCK** - Low Stock,
 - **OUT_OF_STOCK** - Out of Stock,
 - **DISCONTINUED** - Discontinued.
3. Declare the enum type **OrderStatus**, which represents different stages of order fulfillment.
- (a) Add a private, immutable field **description** of type **String**, which stores the description of a given status.
 - (b) Implement the constructor **OrderStatus(String description)**, whose purpose is to initialize the description field.
 - i. Accept the parameter **description**.
 - ii. Assign its value to the field **description**.
 - (c) Add the method **getDescription()**, which returns the value of the field **description**.

- (d) Override the method `toString()` so that it returns a readable representation of the order status.
 - Return the value of the field `description` as the result of the method.
4. Define enum constants corresponding to subsequent order states, assigning each of them a text description passed to the constructor:
 - `NEW` - New Order,
 - `CONFIRMED` - Confirmed,
 - `PROCESSING` - Processing,
 - `SHIPPED` - Shipped,
 - `DELIVERED` - Delivered,
 - `CANCELLED` - Cancelled;
 5. Declare the enum type `Category`, which represents product categories together with business parameters and the individual behavior of each enum constant through overriding abstract methods within the constant definitions.
 - (a) Add private, immutable fields `displayName`, `margin`, `vatRate`, and `defaultWarrantyMonths`, which store, respectively, the display name of the category, the margin, the VAT rate, and the default warranty period in months.
 - (b) Implement the constructor `Category(String displayName, double margin, double vatRate, int defaultWarrantyMonths)`.
 - i. Accept the parameters describing the category.
 - ii. Assign the provided values to the corresponding class fields.
 - (c) Add accessor methods `getDisplayName()`, `getMargin()`, `getVatRate()`, and `getDefaultWarrantyMonths()`, which return the values of the corresponding fields.
 - (d) Declare the abstract method `getDescription()`, whose purpose is to return a detailed textual description specific to a given enum constant.
 - (e) Declare the abstract method `requiresWarranty()`, whose purpose is to determine whether a given category requires a warranty.
 - (f) Declare the abstract method `getDefaultWarrantyPeriod()`, whose purpose is to return the default warranty period appropriate for a given category.
 - (g) Implement the method `calculateSellingPrice(BigDecimal purchasePrice)`.
 - i. Calculate the price increased by the margin by multiplying `purchasePrice` by the value `1 + margin`.
 - ii. Store the result in the helper variable `priceWithMargin`.
 - iii. Calculate the gross price by multiplying `priceWithMargin` by the value `1 + vatRate`.
 - iv. Store the result in the helper variable `grossPrice`.
 - v. Round the result to two decimal places using `RoundingMode.HALF_UP`.

- vi. Return the calculated value.
- (h) Implement the method `calculateVAT(BigDecimal sellingPrice)`.
 - i. Calculate the net price by dividing `sellingPrice` by the value `1 + vatRate`, using scale 2 and `RoundingMode.HALF_UP`.
 - ii. Store the result in the variable `netPrice`.
 - iii. Subtract `netPrice` from `sellingPrice`.
 - iv. Return the resulting value as the VAT amount.
- (i) Implement the method `displayInfo()`.
 - Display the category name, margin percentage, VAT rate percentage, default number of warranty months, and the text returned by `getDescription()`.
 - Use `System.out.printf` and a format consistent with the column layout provided in the code.
- (j) Implement the static method `findByName(String name)`.
 - i. Iterate over all values returned by `values()`.
 - ii. For each category, compare `getDisplayname()` and `name()` with the parameter `name`, using `equalsIgnoreCase()`.
 - iii. Return the first matching category.
 - iv. If no match is found, return `null`.
- (k) Define the enum constant `ELECTRONICS("Electronics", 0.25, 0.23, 24)` and override its abstract methods.
 - i. Implement `getDescription()` so that it returns exactly `"Electronic equipment, computers, smartphones and accessories"`.
 - ii. Implement `requiresWarranty()` so that it returns `true`.
 - iii. Implement `getDefaultWarrantyPeriod()` so that it returns 24.
- (l) Define the enum constant `CLOTHING("Clothing", 0.40, 0.23, 0)` and override its abstract methods.
 - i. Implement `getDescription()` so that it returns exactly `"Women's, men's and children's clothing and accessories"`.
 - ii. Implement `requiresWarranty()` so that it returns `false`.
 - iii. Implement `getDefaultWarrantyPeriod()` so that it returns 0.
- (m) Define the enum constant `HOME("Home & Garden", 0.30, 0.23, 12)` and override its abstract methods.
 - i. Implement `getDescription()` so that it returns exactly `"Furniture, decorations, tools and home equipment"`.
 - ii. Implement `requiresWarranty()` so that it returns `true`.
 - iii. Implement `getDefaultWarrantyPeriod()` so that it returns 12.
- (n) Define the enum constant `SPORTS("Sports & Recreation", 0.35, 0.23, 6)` and override its abstract methods.

- i. Implement `getDescription()` so that it returns exactly "Sports equipment, fitness, tourism and outdoor".
 - ii. Implement `requiresWarranty()` so that it returns `false`.
 - iii. Implement `getDefaultWarrantyPeriod()` so that it returns 6.
- (o) Define the enum constant `BOOKS("Books", 0.20, 0.05, 0)` and override its abstract methods.
 - i. Implement `getDescription()` so that it returns exactly "Literature, guides, scientific and educational books".
 - ii. Implement `requiresWarranty()` so that it returns `false`.
 - iii. Implement `getDefaultWarrantyPeriod()` so that it returns 0.
- (p) Define the enum constant `HEALTH("Health & Beauty", 0.50, 0.08, 0)` and override its abstract methods.
 - i. Implement `getDescription()` so that it returns exactly "Supplements, cosmetics, hygiene and care products".
 - ii. Implement `requiresWarranty()` so that it returns `false`.
 - iii. Implement `getDefaultWarrantyPeriod()` so that it returns 0.

6. Check whether the program output is:

```
=====
ONLINE STORE - VERSION WITHOUT STREAM API AND ICONS
=====

=====
PRODUCT CATEGORIES
=====
Electronics      | Margin: 25.0% | VAT: 23.0% | Warranty: 24 months | Electronic equipment, computers, sma
Clothing         | Margin: 40.0% | VAT: 23.0% | Warranty: 0 months | Women's, men's and children's cloth
Home & Garden    | Margin: 30.0% | VAT: 23.0% | Warranty: 12 months | Furniture, decorations, tools and h
Sports & Recreation | Margin: 35.0% | VAT: 23.0% | Warranty: 6 months | Sports equipment, fitness, tourism a
Books           | Margin: 20.0% | VAT: 5.0% | Warranty: 0 months | Literature, guides, scientific and e
Health & Beauty  | Margin: 50.0% | VAT: 8.0% | Warranty: 0 months | Supplements, cosmetics, hygiene and

Adding products to the store...

=====

=====
ANALYSIS USING ENUMSET
=====

Categories with margin > 30%:
- Clothing          (40%)
- Sports & Recreation (35%)
- Health & Beauty    (50%)
```

Categories requiring warranty:

- Electronics (24 months)
- Home & Garden (12 months)

Categories with reduced VAT rate (<20%):

- Books (5%)
- Health & Beauty (8%)

=====

PRODUCT SEARCH

=====

7. Go to the class `Product` and make sure that it implements the interface `Comparable<Product>`.
 - (a) Go to marker `TODO 01` and implement the method `compareTo(Product other)`, whose purpose is to define the natural ordering of objects of type `Product` based on their names.
 - i. Retrieve the name of the current object from the field `name`.
 - ii. Retrieve the name of the object passed as the parameter `other.name`.
 - iii. Call the method `compareToIgnoreCase(...)` on the name of the current object, passing the name of the object `other` as the argument.
 - iv. Return the comparison result as a value of type `int`, which determines the lexicographic order without regard to letter case.
 - (b) Go to marker `TODO 02` and implement the method `equals(Object o)`, whose purpose is to determine the equality of two objects of type `Product` based on their unique identifier.
 - i. Check whether the current object is the same object as the parameter `o`.
 - If `this == o`, return `true`, because both references point to the same object.
 - ii. Check whether the provided object is `null` or whether its class differs from the class of the current object.
 - If `o == null` or `getClass() != o.getClass()`, return `false`.
 - iii. Cast the object `o` to type `Product` and store the result in the variable `product`.
 - iv. Compare the values of the field `code` of both objects using the method `Objects.equals(...)`.
 - Pass `code` of the current object and `product.code` as the arguments.
 - Return the result of the comparison as the method result.
8. Go to marker `TODO 03` and implement the `hashCode()` method whose purpose is to generate a hash code consistent with the implementation of the `equals(...)` method.
 - (a) Use the `code` field as the basis for calculating the hash value.
 - (b) Call the `Objects.hash(...)` method, passing `code` as an argument.
 - (c) Return the resulting `int` value as the result of the method.

(d) Ensure consistency with the `equals(...)` method.

- Objects considered equal based on the `code` field must return the same hash code.

9. Find all markers `TODO Product` and uncomment the code sections below them. Check whether the program output is:

```
=====
ONLINE STORE - VERSION WITHOUT STREAM API AND ICONS
=====

=====
PRODUCT CATEGORIES
=====
Electronics      | Margin: 25.0% | VAT: 23.0% | Warranty: 24 months | Electronic equipment, computers, sma
Clothing         | Margin: 40.0% | VAT: 23.0% | Warranty: 0 months  | Women's, men's and children's cloth
Home & Garden    | Margin: 30.0% | VAT: 23.0% | Warranty: 12 months | Furniture, decorations, tools and h
Sports & Recreation | Margin: 35.0% | VAT: 23.0% | Warranty: 6 months  | Sports equipment, fitness, tourism a
Books           | Margin: 20.0% | VAT: 5.0%  | Warranty: 0 months  | Literature, guides, scientific and e
Health & Beauty   | Margin: 50.0% | VAT: 8.0%  | Warranty: 0 months  | Supplements, cosmetics, hygiene and

Adding products to the store...

ADDED: Product Dell XPS 15 Laptop
ADDED: Product Samsung Galaxy S23 Smartphone
ADDED: Product Sony WH-1000XM5 Headphones
ADDED: Product LG UltraWide 34" Monitor
ADDED: Product Apple iPad Pro 12.9" Tablet
ADDED: Product The North Face Winter Jacket
ADDED: Product Levi's 501 Jeans
ADDED: Product Nike Sportswear Hoodie
ADDED: Product Adidas Ultraboost Running Shoes
ADDED: Product Ergonomic Office Chair
ADDED: Product LED Floor Lamp
ADDED: Product Wool Rug 200x300cm
ADDED: Product 3-Seater Corner Sofa
ADDED: Product Trek Mountain Bike
ADDED: Product Manduka Yoga Mat
ADDED: Product Adjustable Dumbbells 2x20kg
ADDED: Product Electric Treadmill
ADDED: Product Clean Code - Robert Martin
ADDED: Product Sapiens - Yuval Harari
ADDED: Product The Witcher - Andrzej Sapkowski
ADDED: Product Atomic Habits - James Clear
ADDED: Product Vitamin D3 4000 IU
ADDED: Product Whey Protein WPC 1kg
ADDED: Product Anti-Wrinkle Cream

WARNING: Product with code EL001 already exists!

=====
```

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STORE STATISTICS

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Electronics	Products:	5	Value: \$	74500.00	Profit: \$	32940.51	Revenue: \$	0.00
Clothing	Products:	4	Value: \$	19450.00	Profit: \$	13577.90	Revenue: \$	0.00
Home & Garden	Products:	4	Value: \$	19600.00	Profit: \$	9629.72	Revenue: \$	0.00
Sports & Recreation	Products:	4	Value: \$	21760.00	Profit: \$	12579.20	Revenue: \$	0.00
Books	Products:	4	Value: \$	7300.00	Profit: \$	1898.00	Revenue: \$	0.00
Health & Beauty	Products:	3	Value: \$	5700.00	Profit: \$	3145.20	Revenue: \$	0.00

TOTAL: Products: 24 | Value: \$ 148310.00 | Potential profit: \$ 73770.53

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ANALYSIS USING ENUMSET

=====

Categories with margin > 30%:

- Clothing (40%)
- Sports & Recreation (35%)
- Health & Beauty (50%)

Categories requiring warranty:

- Electronics (24 months)
- Home & Garden (12 months)

Categories with reduced VAT rate (<20%):

- Books (5%)
 - Health & Beauty (8%)
- =====

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PRODUCT SEARCH

=====

Product with code EL002:

- [EL002] Samsung Galaxy S23 Smartphone | \$3874.50 -10% | 12 pcs | Available | Rating: 4.7 (2 reviews)
- =====

ENUMMAP DEMONSTRATION

=====

Number of products in each category:

Electronics	:	5 products
Clothing	:	4 products
Home & Garden	:	4 products
Sports & Recreation	:	4 products

Books : 4 products
Health & Beauty : 3 products

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DEMONSTRATION OF equals(), hashCode() and Comparable

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p1 (EL001): Dell XPS 15 Laptop
p2 (EL002): Samsung Galaxy S23 Smartphone
p3 (EL001): Dell XPS 15 Laptop

p1 == p3: true
p1.equals(p3): true
p1.hashCode(): 66034761
p3.hashCode(): 66034761

p1.compareTo(p2): -15 (alphabetic sorting: Dell XPS 15 Laptop vs Samsung Galaxy S23 Smartphone)

=====

ENUM METHODS DEMONSTRATION

=====

All categories (values()):
- ELECTRONICS -> Electronics
- CLOTHING -> Clothing
- HOME -> Home & Garden
- SPORTS -> Sports & Recreation
- BOOKS -> Books
- HEALTH -> Health & Beauty

Ordinal (position in enum):
Electronics: ordinal = 0
Clothing: ordinal = 1
Home & Garden: ordinal = 2
Sports & Recreation: ordinal = 3
Books: ordinal = 4
Health & Beauty: ordinal = 5

valueOf() - String to Enum conversion:
Category.valueOf("ELECTRONICS") = Electronics
Category.findByName("Books") = Books

=====

ENUM AvailabilityStatus DEMONSTRATION

=====

Quantity 0 pcs -> Out of Stock
Quantity 2 pcs -> Low Stock
Quantity 5 pcs -> Low Stock
Quantity 10 pcs -> Available
Quantity 100 pcs -> Available

=====

ENUM DiscountType DEMONSTRATION

=====

Base price: \$1000

No discount	-> \$1000.00
-5%	-> \$950.00
-10%	-> \$900.00
-15%	-> \$850.00
-20%	-> \$800.00
-30%	-> \$700.00
-50%	-> \$500.00
-70%	-> \$300.00

10. 🚩 Milestone 1: **present the result of your work to the instructor**
11. Declare the class **ProductComparators**, which will provide a set of static methods returning different implementations of the interface **Comparator<Product>** for sorting **Product** objects according to different criteria.
 - (a) Implement the method **byName()**, which returns a comparator comparing products by name.
 - Compare the values **p1.getName()** and **p2.getName()** using **compareToIgnoreCase()**.
 - (b) Implement the method **byCode()**, which returns a comparator comparing products by code.
 - Compare the values **p1.getCode()** and **p2.getCode()** using **compareTo()**.
 - (c) Implement the method **byPriceAscending()**, which sorts products in ascending order by the price after discount.
 - Compare the values **p1.getPriceAfterDiscount()** and **p2.getPriceAfterDiscount()**.
 - (d) Implement the method **byPriceDescending()**, which sorts products in descending order by the price after discount.
 - Compare the values in reverse order by calling **p2.getPriceAfterDiscount().compareTo(p1.getPriceAfterDiscount())**.
 - (e) Implement the method **byCategoryAndPrice()**, which sorts products first by category and then by price.
 - i. Compare the product categories using **p1.getCategory().compareTo(p2.getCategory())**.
 - ii. If the result is not zero, return it.
 - iii. Otherwise, compare the discounted prices.
 - (f) Implement the method **byStockQuantity()**, which sorts products in descending order by the number of available units.
 - Use **Integer.compare(p2.getStockQuantity(), p1.getStockQuantity())**.
 - (g) Implement the method **byProfit()**, which sorts products in descending order by potential profit.

- Compare the values `calculatePotentialProfit()` in reverse order.
- (h) Implement the method `byManufacturer()`, which sorts products by manufacturer and then by name.
- i. Compare manufacturers using `compareToIgnoreCase()`.
 - ii. If the result is not zero, return it.
 - iii. Otherwise, compare the product names.
- (i) Implement the method `byRating()`, which sorts products by rating and then by review count.
- i. Compare ratings in descending order using `Double.compare(p2.getRating(), p1.getRating())`.
 - ii. If the result is not zero, return it.
 - iii. Otherwise, compare review counts in descending order.
- (j) Implement the method `byDaysInStock()`, which sorts products in descending order by the number of days in stock.
- Use `Integer.compare(p2.getDaysInStock(), p1.getDaysInStock())`.
- (k) Implement the method `byStatus()`, which sorts products by availability status and then by stock quantity.
- i. Compare the statuses using `compareTo()`.
 - ii. If the result is not zero, return it.
 - iii. Otherwise, compare stock quantities in ascending order.
- (l) Implement the method `multiCriteria()`, which sorts products using multiple criteria in a fixed order.
- i. Compare product categories.
 - ii. If the result is not zero, return it.
 - iii. Compare ratings in descending order.
 - iv. If the result is not zero, return it.
 - v. Compare prices in ascending order.
 - vi. If the result is not zero, return it.
 - vii. Compare product names ignoring case and return the result.

12. Find all markers `TODO ProductComparators` and uncomment the code sections below them. Check whether the program output is:

```
=====
ONLINE STORE - VERSION WITHOUT STREAM API AND ICONS
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Home & Garden		Margin: 30.0%		VAT: 23.0%		Warranty: 12 months		Furniture, decorations, tools and h
Sports & Recreation		Margin: 35.0%		VAT: 23.0%		Warranty: 6 months		Sports equipment, fitness, tourism a
Books		Margin: 20.0%		VAT: 5.0%		Warranty: 0 months		Literature, guides, scientific and e
Health & Beauty		Margin: 50.0%		VAT: 8.0%		Warranty: 0 months		Supplements, cosmetics, hygiene and

Adding products to the store...

ADDED: Product Dell XPS 15 Laptop
 ADDED: Product Samsung Galaxy S23 Smartphone
 ADDED: Product Sony WH-1000XM5 Headphones
 ADDED: Product LG UltraWide 34" Monitor
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 ADDED: Product The North Face Winter Jacket
 ADDED: Product Levi's 501 Jeans
 ADDED: Product Nike Sportswear Hoodie
 ADDED: Product Adidas Ultraboost Running Shoes
 ADDED: Product Ergonomic Office Chair
 ADDED: Product LED Floor Lamp
 ADDED: Product Wool Rug 200x300cm
 ADDED: Product 3-Seater Corner Sofa
 ADDED: Product Trek Mountain Bike
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 ADDED: Product Vitamin D3 4000 IU
 ADDED: Product Whey Protein WPC 1kg
 ADDED: Product Anti-Wrinkle Cream

WARNING: Product with code EL001 already exists!

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STORE STATISTICS

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Clothing		Products:	4		Value: \$	19450.00		Profit: \$	13577.90		Revenue: \$	0.00
Home & Garden		Products:	4		Value: \$	19600.00		Profit: \$	9629.72		Revenue: \$	0.00
Sports & Recreation		Products:	4		Value: \$	21760.00		Profit: \$	12579.20		Revenue: \$	0.00
Books		Products:	4		Value: \$	7300.00		Profit: \$	1898.00		Revenue: \$	0.00
Health & Beauty		Products:	3		Value: \$	5700.00		Profit: \$	3145.20		Revenue: \$	0.00

TOTAL:		Products:	24		Value: \$	148310.00		Potential profit: \$	73770.53
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PROMOTIONS

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[EL005] Apple iPad Pro 12.9" Tablet	\$5488.88 -15% 2 pcs Low Stock Rating: 5.0 (2 reviews)
[EL002] Samsung Galaxy S23 Smartphone	\$3874.50 -10% 12 pcs Available Rating: 4.7 (2 reviews)
[HM004] 3-Seater Corner Sofa	\$2814.24 -20% 3 pcs Low Stock Rating: 4.4 (1 reviews)
[SP004] Electric Treadmill	\$2092.23 -30% 2 pcs Low Stock Rating: 4.7 (1 reviews)
[CL003] Nike Sportswear Hoodie	\$294.46 -5% 30 pcs Available Rating: 4.4 (1 reviews)
[HL002] Whey Protein WPC 1kg	\$116.64 -10% 30 pcs Available Rating: 4.6 (1 reviews)

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TOP 5 PRODUCTS

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1. [BK001] Clean Code - Robert Martin	\$81.90 40 pcs Available Rating: 5.0 (3 reviews)
2. [EL005] Apple iPad Pro 12.9" Tablet	\$5488.88 -15% 2 pcs Low Stock Rating: 5.0 (2 reviews)
3. [EL003] Sony WH-1000XM5 Headphones	\$1845.00 8 pcs Available Rating: 4.9 (4 reviews)
4. [SP001] Trek Mountain Bike	\$4151.25 4 pcs Low Stock Rating: 4.9 (1 reviews)
5. [BK003] The Witcher - Andrzej Sapkowski	\$44.10 50 pcs Available Rating: 4.9 (1 reviews)

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PRODUCTS TO REORDER (stock <= 5 pcs)

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[EL001] Dell XPS 15 Laptop	\$5381.25 5 pcs Low Stock Rating: 4.9 (3 reviews)
[HM003] Wool Rug 200x300cm	\$1279.20 5 pcs Low Stock Rating: 4.6 (1 reviews)
[SP001] Trek Mountain Bike	\$4151.25 4 pcs Low Stock Rating: 4.9 (1 reviews)
[CL004] Adidas Ultraboost Running Shoes	\$774.90 4 pcs Low Stock Rating: 4.9 (2 reviews)
[EL004] LG UltraWide 34" Monitor	\$2767.50 3 pcs Low Stock Rating: 4.5 (1 reviews)
[HM004] 3-Seater Corner Sofa	\$2814.24 -20% 3 pcs Low Stock Rating: 4.4 (1 reviews)
[SP004] Electric Treadmill	\$2092.23 -30% 2 pcs Low Stock Rating: 4.7 (1 reviews)
[EL005] Apple iPad Pro 12.9" Tablet	\$5488.88 -15% 2 pcs Low Stock Rating: 5.0 (2 reviews)

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CATEGORY: ELECTRONICS

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[EL005] Apple iPad Pro 12.9" Tablet	\$5488.88 -15% 2 pcs Low Stock Rating: 5.0 (2 reviews)
[EL001] Dell XPS 15 Laptop	\$5381.25 5 pcs Low Stock Rating: 4.9 (3 reviews)
[EL002] Samsung Galaxy S23 Smartphone	\$3874.50 -10% 12 pcs Available Rating: 4.7 (2 reviews)
[EL004] LG UltraWide 34" Monitor	\$2767.50 3 pcs Low Stock Rating: 4.5 (1 reviews)
[EL003] Sony WH-1000XM5 Headphones	\$1845.00 8 pcs Available Rating: 4.9 (4 reviews)

=====

Number of products: 5

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CATEGORY: BOOKS

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[BK001]	Clean Code - Robert Martin		\$81.90		40 pcs		Available		Rating: 5.0 (3 reviews)
[BK003]	The Witcher - Andrzej Sapkowski		\$44.10		50 pcs		Available		Rating: 4.9 (1 reviews)
[BK002]	Sapiens - Yuval Harari		\$56.70		35 pcs		Available		Rating: 4.9 (2 reviews)
[BK004]	Atomic Habits - James Clear		\$69.30		25 pcs		Available		Rating: 4.8 (2 reviews)

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Number of products: 4

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ALL PRODUCTS

Sorting: by rating (best first)

=====

[BK001]	Clean Code - Robert Martin		\$81.90		40 pcs		Available		Rating: 5.0 (3 reviews)
[EL005]	Apple iPad Pro 12.9" Tablet		\$5488.88		-15%		2 pcs		Low Stock Rating: 5.0 (2 reviews)
[EL003]	Sony WH-1000XM5 Headphones		\$1845.00		8 pcs		Available		Rating: 4.9 (4 reviews)
[SP001]	Trek Mountain Bike		\$4151.25		4 pcs		Low Stock		Rating: 4.9 (1 reviews)
[BK003]	The Witcher - Andrzej Sapkowski		\$44.10		50 pcs		Available		Rating: 4.9 (1 reviews)
[EL001]	Dell XPS 15 Laptop		\$5381.25		5 pcs		Low Stock		Rating: 4.9 (3 reviews)
[BK002]	Sapiens - Yuval Harari		\$56.70		35 pcs		Available		Rating: 4.9 (2 reviews)
[CL004]	Adidas Ultraboost Running Shoes		\$774.90		4 pcs		Low Stock		Rating: 4.9 (2 reviews)
[BK004]	Atomic Habits - James Clear		\$69.30		25 pcs		Available		Rating: 4.8 (2 reviews)
[SP004]	Electric Treadmill		\$2092.23		-30%		2 pcs		Low Stock Rating: 4.7 (1 reviews)
[CL002]	Levi's 501 Jeans		\$430.50		25 pcs		Available		Rating: 4.7 (1 reviews)
[EL002]	Samsung Galaxy S23 Smartphone		\$3874.50		-10%		12 pcs		Available Rating: 4.7 (2 reviews)
[HL002]	Whey Protein WPC 1kg		\$116.64		-10%		30 pcs		Available Rating: 4.6 (1 reviews)
[HM003]	Wool Rug 200x300cm		\$1279.20		5 pcs		Low Stock		Rating: 4.6 (1 reviews)
[SP002]	Manduka Yoga Mat		\$365.31		18 pcs		Available		Rating: 4.5 (1 reviews)
[EL004]	LG UltraWide 34" Monitor		\$2767.50		3 pcs		Low Stock		Rating: 4.5 (1 reviews)
[HL001]	Vitamin D3 4000 IU		\$40.50		60 pcs		Available		Rating: 4.5 (1 reviews)
[CL001]	The North Face Winter Jacket		\$688.80		15 pcs		Available		Rating: 4.4 (2 reviews)
[HM004]	3-Seater Corner Sofa		\$2814.24		-20%		3 pcs		Low Stock Rating: 4.4 (1 reviews)
[CL003]	Nike Sportswear Hoodie		\$294.46		-5%		30 pcs		Available Rating: 4.4 (1 reviews)
[SP003]	Adjustable Dumbbells 2x20kg		\$581.18		12 pcs		Available		Rating: 4.3 (1 reviews)
[HM001]	Ergonomic Office Chair		\$959.40		10 pcs		Available		Rating: 4.2 (1 reviews)
[HL003]	Anti-Wrinkle Cream		\$194.40		15 pcs		Available		Rating: 4.2 (1 reviews)
[HM002]	LED Floor Lamp		\$239.85		20 pcs		Available		Rating: 4.0 (1 reviews)

=====

Total products: 24

=====

=====

ALL PRODUCTS

Sorting: multi-criteria: category -> rating -> price -> name

```

=====
[EL005] Apple iPad Pro 12.9" Tablet | $5488.88 -15% | 2 pcs | Low Stock | Rating: 5.0 (2 reviews)
[EL003] Sony WH-1000XM5 Headphones | $1845.00 | 8 pcs | Available | Rating: 4.9 (4 reviews)
[EL001] Dell XPS 15 Laptop | $5381.25 | 5 pcs | Low Stock | Rating: 4.9 (3 reviews)
[EL002] Samsung Galaxy S23 Smartphone | $3874.50 -10% | 12 pcs | Available | Rating: 4.7 (2 reviews)
[EL004] LG UltraWide 34" Monitor | $2767.50 | 3 pcs | Low Stock | Rating: 4.5 (1 reviews)
[CL004] Adidas Ultraboost Running Shoes | $774.90 | 4 pcs | Low Stock | Rating: 4.9 (2 reviews)
[CL002] Levi's 501 Jeans | $430.50 | 25 pcs | Available | Rating: 4.7 (1 reviews)
[CL003] Nike Sportswear Hoodie | $294.46 -5% | 30 pcs | Available | Rating: 4.4 (1 reviews)
[CL001] The North Face Winter Jacket | $688.80 | 15 pcs | Available | Rating: 4.4 (2 reviews)
[HM003] Wool Rug 200x300cm | $1279.20 | 5 pcs | Low Stock | Rating: 4.6 (1 reviews)
[HM004] 3-Seater Corner Sofa | $2814.24 -20% | 3 pcs | Low Stock | Rating: 4.4 (1 reviews)
[HM001] Ergonomic Office Chair | $959.40 | 10 pcs | Available | Rating: 4.2 (1 reviews)
[HM002] LED Floor Lamp | $239.85 | 20 pcs | Available | Rating: 4.0 (1 reviews)
[SP001] Trek Mountain Bike | $4151.25 | 4 pcs | Low Stock | Rating: 4.9 (1 reviews)
[SP004] Electric Treadmill | $2092.23 -30% | 2 pcs | Low Stock | Rating: 4.7 (1 reviews)
[SP002] Manduka Yoga Mat | $365.31 | 18 pcs | Available | Rating: 4.5 (1 reviews)
[SP003] Adjustable Dumbbells 2x20kg | $581.18 | 12 pcs | Available | Rating: 4.3 (1 reviews)
[BK001] Clean Code - Robert Martin | $81.90 | 40 pcs | Available | Rating: 5.0 (3 reviews)
[BK003] The Witcher - Andrzej Sapkowski | $44.10 | 50 pcs | Available | Rating: 4.9 (1 reviews)
[BK002] Sapiens - Yuval Harari | $56.70 | 35 pcs | Available | Rating: 4.9 (2 reviews)
[BK004] Atomic Habits - James Clear | $69.30 | 25 pcs | Available | Rating: 4.8 (2 reviews)
[HL002] Whey Protein WPC 1kg | $116.64 -10% | 30 pcs | Available | Rating: 4.6 (1 reviews)
[HL001] Vitamin D3 4000 IU | $40.50 | 60 pcs | Available | Rating: 4.5 (1 reviews)
[HL003] Anti-Wrinkle Cream | $194.40 | 15 pcs | Available | Rating: 4.2 (1 reviews)
=====

```

Total products: 24

ALL PRODUCTS

Sorting: by potential profit (highest first)

```

=====
[EL002] Samsung Galaxy S23 Smartphone | $3874.50 -10% | 12 pcs | Available | Rating: 4.7 (2 reviews)
[EL001] Dell XPS 15 Laptop | $5381.25 | 5 pcs | Low Stock | Rating: 4.9 (3 reviews)
[SP001] Trek Mountain Bike | $4151.25 | 4 pcs | Low Stock | Rating: 4.9 (1 reviews)
[EL003] Sony WH-1000XM5 Headphones | $1845.00 | 8 pcs | Available | Rating: 4.9 (4 reviews)
[CL002] Levi's 501 Jeans | $430.50 | 25 pcs | Available | Rating: 4.7 (1 reviews)
[CL001] The North Face Winter Jacket | $688.80 | 15 pcs | Available | Rating: 4.4 (2 reviews)
[HM001] Ergonomic Office Chair | $959.40 | 10 pcs | Available | Rating: 4.2 (1 reviews)
[CL003] Nike Sportswear Hoodie | $294.46 -5% | 30 pcs | Available | Rating: 4.4 (1 reviews)
[EL004] LG UltraWide 34" Monitor | $2767.50 | 3 pcs | Low Stock | Rating: 4.5 (1 reviews)
[SP003] Adjustable Dumbbells 2x20kg | $581.18 | 12 pcs | Available | Rating: 4.3 (1 reviews)
[SP002] Manduka Yoga Mat | $365.31 | 18 pcs | Available | Rating: 4.5 (1 reviews)
[EL005] Apple iPad Pro 12.9" Tablet | $5488.88 -15% | 2 pcs | Low Stock | Rating: 5.0 (2 reviews)
[HM003] Wool Rug 200x300cm | $1279.20 | 5 pcs | Low Stock | Rating: 4.6 (1 reviews)
[HM004] 3-Seater Corner Sofa | $2814.24 -20% | 3 pcs | Low Stock | Rating: 4.4 (1 reviews)
[HM002] LED Floor Lamp | $239.85 | 20 pcs | Available | Rating: 4.0 (1 reviews)
[CL004] Adidas Ultraboost Running Shoes | $774.90 | 4 pcs | Low Stock | Rating: 4.9 (2 reviews)
[HL003] Anti-Wrinkle Cream | $194.40 | 15 pcs | Available | Rating: 4.2 (1 reviews)
[HL002] Whey Protein WPC 1kg | $116.64 -10% | 30 pcs | Available | Rating: 4.6 (1 reviews)

```

[HL001]	Vitamin D3 4000 IU	\$40.50 60 pcs Available Rating: 4.5 (1 reviews)
[BK001]	Clean Code - Robert Martin	\$81.90 40 pcs Available Rating: 5.0 (3 reviews)
[SP004]	Electric Treadmill	\$2092.23 -30% 2 pcs Low Stock Rating: 4.7 (1 reviews)
[BK003]	The Witcher - Andrzej Sapkowski	\$44.10 50 pcs Available Rating: 4.9 (1 reviews)
[BK002]	Sapiens - Yuval Harari	\$56.70 35 pcs Available Rating: 4.9 (2 reviews)
[BK004]	Atomic Habits - James Clear	\$69.30 25 pcs Available Rating: 4.8 (2 reviews)

=====

Total products: 24

=====

SELECTED PRODUCT DETAILS

=====

=====

PRODUCT DETAILS - Electronics

=====

Code:	EL001
Name:	Dell XPS 15 Laptop
Manufacturer:	Dell
Category:	Electronics

Purchase price:	\$3500.00
Selling price:	\$5381.25
VAT (23%):	\$1006.25

Stock quantity:	5 pcs
Status:	Low Stock
Stock value:	\$17500.00
Potential profit:	\$9406.25

Delivery date:	04/08/2026
Days in stock:	0
Warranty:	24 months

Rating:	4.9 / 5.0
Review count:	3

=====

=====

PRODUCT DETAILS - Electronics

=====

Code:	EL002
Name:	Samsung Galaxy S23 Smartphone
Manufacturer:	Samsung
Category:	Electronics

Purchase price:	\$2800.00
Selling price:	\$4305.00
Discount:	-10%
Price after discount:	\$3874.50
VAT (23%):	\$724.50

Stock quantity: 12 pcs
Status: Available
Stock value: \$33600.00
Potential profit: \$12894.00

Delivery date: 04/08/2026
Days in stock: 0
Warranty: 24 months

Rating: 4.7 / 5.0
Review count: 2
=====

=====

=====

ANALYSIS USING ENUMSET

=====

Categories with margin > 30%:
- Clothing (40%)
- Sports & Recreation (35%)
- Health & Beauty (50%)

Categories requiring warranty:
- Electronics (24 months)
- Home & Garden (12 months)

Categories with reduced VAT rate (<20%):
- Books (5%)
- Health & Beauty (8%)
=====

=====

PRODUCT SEARCH

=====

Sony products:
- [EL003] Sony WH-1000XM5 Headphones | \$1845.00 | 8 pcs | Available | Rating: 4.9 (4 reviews)

Product with code EL002:
- [EL002] Samsung Galaxy S23 Smartphone | \$3874.50 -10% | 12 pcs | Available | Rating: 4.7 (2 reviews)

=====

ENUMMAP DEMONSTRATION

=====

Number of products in each category:
Electronics : 5 products
Clothing : 4 products
Home & Garden : 4 products

```

Sports & Recreation : 4 products
Books                : 4 products
Health & Beauty     : 3 products

=====
DEMONSTRATION OF equals(), hashCode() and Comparable
=====

p1 (EL001): Dell XPS 15 Laptop
p2 (EL002): Samsung Galaxy S23 Smartphone
p3 (EL001): Dell XPS 15 Laptop

p1 == p3:          true
p1.equals(p3):     true
p1.hashCode():     66034761
p3.hashCode():     66034761

p1.compareTo(p2):  -15 (alphabetic sorting: Dell XPS 15 Laptop vs Samsung Galaxy S23 Smartphone)

=====
ENUM METHODS DEMONSTRATION
=====

All categories (values()):
- ELECTRONICS -> Electronics
- CLOTHING -> Clothing
- HOME -> Home & Garden
- SPORTS -> Sports & Recreation
- BOOKS -> Books
- HEALTH -> Health & Beauty

Ordinal (position in enum):
Electronics: ordinal = 0
Clothing: ordinal = 1
Home & Garden: ordinal = 2
Sports & Recreation: ordinal = 3
Books: ordinal = 4
Health & Beauty: ordinal = 5

valueOf() - String to Enum conversion:
Category.valueOf("ELECTRONICS") = Electronics
Category.findByName("Books") = Books

=====
ENUM AvailabilityStatus DEMONSTRATION
=====

Quantity 0 pcs -> Out of Stock
Quantity 2 pcs -> Low Stock
Quantity 5 pcs -> Low Stock
Quantity 10 pcs -> Available
Quantity 100 pcs -> Available

```

```
=====
ENUM DiscountType DEMONSTRATION
=====
```

Base price: \$1000

No discount	-> \$1000.00
-5%	-> \$950.00
-10%	-> \$900.00
-15%	-> \$850.00
-20%	-> \$800.00
-30%	-> \$700.00
-50%	-> \$500.00
-70%	-> \$300.00

13. Declare the class `OrderItem`, which represents a single order line containing a product, its quantity, and the unit price.
 - (a) Add a private, immutable field `product` of type `Product`, which stores a reference to the product assigned to the line item.
 - (b) Add a private, immutable field `quantity` of type `int`, which specifies the number of units of the given product in this line item.
 - (c) Add a private, immutable field `unitPrice` of type `BigDecimal`, which stores the unit price of the product after discount at the moment it is added to the order.
 - (d) Implement the constructor `OrderItem(Product product, int quantity)`, whose purpose is to initialize all class fields.
 - i. Accept the parameter `product`, representing the product.
 - ii. Accept the parameter `quantity`, representing the number of units.
 - iii. Assign the provided product to the field `product`.
 - iv. Assign the provided quantity to the field `quantity`.
 - v. Retrieve the current discounted product price by calling `product.getPriceAfterDiscount()`.
 - vi. Assign this value to the field `unitPrice` in order to preserve the price at the time the line item is created.
 - (e) Add accessor methods `getProduct()`, `getQuantity()`, and `getUnitPrice()`, which return the values of the corresponding fields.
 - (f) Implement the method `calculateLineTotal()`, whose purpose is to calculate the total value of this order line.
 - i. Multiply the value `unitPrice` by the number of units `quantity`, using `BigDecimal.valueOf(quantity)`.
 - ii. Apply the method `setScale(2, RoundingMode.HALF_UP)` to obtain a result rounded to two decimal places.
 - iii. Return the calculated value as the method result.

14. Declare the class `Order`, which represents a customer order, implements the interface `Comparable<Order>`, and stores the order number, customer data, a list of line items, the creation date, and the current fulfillment status.
 - (a) Add a private static field `orderCounter` of type `int`, which will store the global counter used to assign successive order numbers.
 - (b) Add a private, immutable field `orderNumber` of type `int`, which stores the number of the current order.
 - (c) Add a private, immutable field `customerName` of type `String`, which stores the name of the customer placing the order.
 - (d) Add a private, immutable field `items` of type `List<OrderItem>`, which stores the list of order line items.
 - (e) Add a private, immutable field `orderDate` of type `LocalDate`, which stores the order creation date.
 - (f) Add a private field `status` of type `OrderStatus`, which stores the current order fulfillment state and may be modified after the object is created.
 - (g) Implement the constructor `Order(String customerName)`, whose purpose is to create a new order with an automatically assigned number, an empty item list, the current date, and the initial status.
 - i. Assign to the field `orderNumber` the current value of the field `orderCounter`, then increment `orderCounter` by one using post-incrementation.
 - ii. Assign the provided parameter `customerName` to the field `customerName`.
 - iii. Create a new instance of `ArrayList<>` and assign it to the field `items`.
 - iv. Set the field `orderDate` to the current date by calling `LocalDate.now()`.
 - v. Set the field `status` to the initial value `OrderStatus.NEW`.
 - (h) Add accessor methods `getOrderNumber()`, `getCustomerName()`, `getItems()`, `getOrderDate()`, and `getStatus()`, which return the values of the corresponding fields.
 - (i) Add the method `setStatus(OrderStatus status)`, whose purpose is to update the field `status` based on the provided argument.
15. Implement the logic for managing order items and calculating the total order value.
 - (a) Implement the method `addItem(Product product, int quantity)`, whose purpose is to add a new line item to the order and simultaneously reduce the product stock level.
 - i. Retrieve the current product stock level by calling `product.getStockQuantity()`.
 - ii. Verify that the available number of units is not smaller than the requested value `quantity`.
 - iii. If the stock level is insufficient, throw an `IllegalStateException` with a message stating that there are too few products in stock.

- iv. If validation succeeds, create a new object of type `OrderItem`, passing `product` and `quantity`.
 - v. Add the created line item to the list `items` by calling `items.add(...)`.
 - vi. Call the method `product.sell(quantity)` to reflect the sale of the corresponding number of units.
- (b) Implement the method `calculateTotalValue()`, whose purpose is to calculate the total value of the entire order without using streams.
- i. Initialize the variable `total` with the value `BigDecimal.ZERO`.
 - ii. Iterate through all elements of the list `items` using a `for-each` loop.
 - iii. For each line item, call the method `item.calculateLineTotal()`.
 - iv. Add the obtained value to the variable `total` by calling `total.add(...)` and assign the result back to `total`.
 - v. Return the calculated value `total`.
- (c) Implement the method `getItemCount()`, whose purpose is to return the number of line items in the order.
- Call the method `items.size()`.
 - Return the obtained value as the number of order lines.
- (d) Implement the method `getTotalProductCount()`, whose purpose is to calculate the total number of product units across all line items in the order without using streams.
- i. Initialize the variable `total` with the value 0.
 - ii. Iterate through all line items stored in the list `items`.
 - iii. For each line item, retrieve the number of units by calling `item.getQuantity()`.
 - iv. Add the obtained value to the variable `total`.
 - v. Return the final value of the variable `total`.
- (e) Implement the method `getCategoryStatistics()`, whose purpose is to build a map containing statistics of the number of sold units grouped by product category.
- i. Create a new map `EnumMap<Category, Integer>` and assign it to the variable `statistics`, passing `Category.class` to the constructor.
 - ii. Iterate through all line items stored in the list `items`.
 - iii. For each line item, retrieve the product category by calling `item.getProduct().getCategory()` and store it in the variable `cat`.
 - iv. Read the current value from the map for the key `cat` by calling `statistics.get(cat)` and store it in the variable `current`.
 - v. Check whether the variable `current` is equal to `null`.
 - A. If no entry exists yet for a given category, add a new key value pair to the map, where the value is `item.getQuantity()`.
 - B. If an entry already exists, update the map by storing under the same key the sum `current + item.getQuantity()`.
 - vi. Return the populated map `statistics`.

16. Implement the methods responsible for order comparison and textual representation.
- (a) Override the method `compareTo(Order other)`, whose purpose is to define the natural ordering of `Order` objects based on the order date.
 - Call `this.orderDate.compareTo(other.orderDate)`.
 - Return the comparison result as a value of type `int`.
 - (b) Override the method `toString()` so that it returns a short, formatted summary of the order.
 - Use `String.format`.
 - Include the order number formatted to four digits in the result.
 - Include the customer name from the field `customerName`.
 - Format the order date by calling `orderDate.format(DateTimeFormatter.ofPattern("MM/dd/yyyy"))`.
 - Include the total order value by calling `calculateTotalValue()`.
 - Include the number of line items by calling `items.size()`.
 - Include the status description by calling `status.getDescription()`.
 - Return the created string.
17. Implement the method that presents full order details in the form of a text report.
- (a) Add the method `displayDetails()`, whose purpose is to display a complete summary of the order data together with the list of line items and a final summary.
 - i. Create an object of type `DateTimeFormatter` with the pattern `"MM/dd/yyyy"` and store it in the variable `formatter`.
 - ii. Display the top separator line consisting of `=` characters using `System.out.println`.
 - iii. Display the order header in the format `ORDER #0000`, using `System.out.printf` and the number `orderNumber`.
 - iv. Display a second separator consisting of `=` characters.
 - v. Display the line containing the customer name using the field `customerName`.
 - vi. Display the line containing the order date by formatting `orderDate` using the prepared object `formatter`.
 - vii. Display the line containing the order status using `status.getDescription()`.
 - viii. Display the middle separator consisting of `-` characters.
 - ix. Iterate through all line items stored in the list `items`.
 - A. For each line item, display the product name and the number of units using `item.getProduct().getName()` and `item.getQuantity()`.
 - B. Then display the line containing the discounted unit price, the number of units, and the line total, using `item.getProduct().getPriceAfterDiscount()`, `item.getQuantity()`, and `item.calculateLineTotal()`, respectively.
 - x. After all line items have been displayed, show another separator consisting of `-` characters.

- xi. Display the line containing the total order value by calling `calculateTotalValue()`.
- xii. Display the final separator consisting of `=` characters.

18. Find all markers `TODO Order` and uncomment the code sections below them. Check whether the program output is:

```
=====
ONLINE STORE - VERSION WITHOUT STREAM API AND ICONS
=====

=====
PRODUCT CATEGORIES
=====
Electronics      | Margin: 25.0% | VAT: 23.0% | Warranty: 24 months | Electronic equipment, computers, sma
Clothing         | Margin: 40.0% | VAT: 23.0% | Warranty: 0 months | Women's, men's and children's cloth
Home & Garden    | Margin: 30.0% | VAT: 23.0% | Warranty: 12 months | Furniture, decorations, tools and ho
Sports & Recreation | Margin: 35.0% | VAT: 23.0% | Warranty: 6 months | Sports equipment, fitness, tourism a
Books            | Margin: 20.0% | VAT: 5.0% | Warranty: 0 months | Literature, guides, scientific and e
Health & Beauty   | Margin: 50.0% | VAT: 8.0% | Warranty: 0 months | Supplements, cosmetics, hygiene and

Adding products to the store...

ADDED: Product Dell XPS 15 Laptop
ADDED: Product Samsung Galaxy S23 Smartphone
ADDED: Product Sony WH-1000XM5 Headphones
ADDED: Product LG UltraWide 34" Monitor
ADDED: Product Apple iPad Pro 12.9" Tablet
ADDED: Product The North Face Winter Jacket
ADDED: Product Levi's 501 Jeans
ADDED: Product Nike Sportswear Hoodie
ADDED: Product Adidas Ultraboost Running Shoes
ADDED: Product Ergonomic Office Chair
ADDED: Product LED Floor Lamp
ADDED: Product Wool Rug 200x300cm
ADDED: Product 3-Seater Corner Sofa
ADDED: Product Trek Mountain Bike
ADDED: Product Manduka Yoga Mat
ADDED: Product Adjustable Dumbbells 2x20kg
ADDED: Product Electric Treadmill
ADDED: Product Clean Code - Robert Martin
ADDED: Product Sapiens - Yuval Harari
ADDED: Product The Witcher - Andrzej Sapkowski
ADDED: Product Atomic Habits - James Clear
ADDED: Product Vitamin D3 4000 IU
ADDED: Product Whey Protein WPC 1kg
ADDED: Product Anti-Wrinkle Cream

WARNING: Product with code EL001 already exists!

=====
```

STORE STATISTICS

Electronics	Products:	5	Value: \$	74500.00	Profit: \$	32940.51	Revenue: \$	0.00
Clothing	Products:	4	Value: \$	19450.00	Profit: \$	13577.90	Revenue: \$	0.00
Home & Garden	Products:	4	Value: \$	19600.00	Profit: \$	9629.72	Revenue: \$	0.00
Sports & Recreation	Products:	4	Value: \$	21760.00	Profit: \$	12579.20	Revenue: \$	0.00
Books	Products:	4	Value: \$	7300.00	Profit: \$	1898.00	Revenue: \$	0.00
Health & Beauty	Products:	3	Value: \$	5700.00	Profit: \$	3145.20	Revenue: \$	0.00

TOTAL: Products: 24 | Value: \$ 148310.00 | Potential profit: \$ 73770.53

PROMOTIONS

[EL005] Apple iPad Pro 12.9" Tablet	\$5488.88 -15% 2 pcs Low Stock Rating: 5.0 (2 reviews)
[EL002] Samsung Galaxy S23 Smartphone	\$3874.50 -10% 12 pcs Available Rating: 4.7 (2 reviews)
[HM004] 3-Seater Corner Sofa	\$2814.24 -20% 3 pcs Low Stock Rating: 4.4 (1 reviews)
[SP004] Electric Treadmill	\$2092.23 -30% 2 pcs Low Stock Rating: 4.7 (1 reviews)
[CL003] Nike Sportswear Hoodie	\$294.46 -5% 30 pcs Available Rating: 4.4 (1 reviews)
[HL002] Whey Protein WPC 1kg	\$116.64 -10% 30 pcs Available Rating: 4.6 (1 reviews)

TOP 5 PRODUCTS

- [BK001] Clean Code - Robert Martin | \$81.90 | 40 pcs | Available | Rating: 5.0 (3 reviews)
- [EL005] Apple iPad Pro 12.9" Tablet | \$5488.88 -15% | 2 pcs | Low Stock | Rating: 5.0 (2 reviews)
- [EL003] Sony WH-1000XM5 Headphones | \$1845.00 | 8 pcs | Available | Rating: 4.9 (4 reviews)
- [SP001] Trek Mountain Bike | \$4151.25 | 4 pcs | Low Stock | Rating: 4.9 (1 reviews)
- [BK003] The Witcher - Andrzej Sapkowski | \$44.10 | 50 pcs | Available | Rating: 4.9 (1 reviews)

PRODUCTS TO REORDER (stock <= 5 pcs)

[EL001] Dell XPS 15 Laptop	\$5381.25 5 pcs Low Stock Rating: 4.9 (3 reviews)
[HM003] Wool Rug 200x300cm	\$1279.20 5 pcs Low Stock Rating: 4.6 (1 reviews)
[SP001] Trek Mountain Bike	\$4151.25 4 pcs Low Stock Rating: 4.9 (1 reviews)
[CL004] Adidas Ultraboost Running Shoes	\$774.90 4 pcs Low Stock Rating: 4.9 (2 reviews)
[EL004] LG UltraWide 34" Monitor	\$2767.50 3 pcs Low Stock Rating: 4.5 (1 reviews)
[HM004] 3-Seater Corner Sofa	\$2814.24 -20% 3 pcs Low Stock Rating: 4.4 (1 reviews)
[SP004] Electric Treadmill	\$2092.23 -30% 2 pcs Low Stock Rating: 4.7 (1 reviews)
[EL005] Apple iPad Pro 12.9" Tablet	\$5488.88 -15% 2 pcs Low Stock Rating: 5.0 (2 reviews)

=====

CATEGORY: ELECTRONICS

=====

[EL005] Apple iPad Pro 12.9" Tablet	\$5488.88 -15% 2 pcs Low Stock Rating: 5.0 (2 reviews)
[EL001] Dell XPS 15 Laptop	\$5381.25 5 pcs Low Stock Rating: 4.9 (3 reviews)
[EL002] Samsung Galaxy S23 Smartphone	\$3874.50 -10% 12 pcs Available Rating: 4.7 (2 reviews)
[EL004] LG UltraWide 34" Monitor	\$2767.50 3 pcs Low Stock Rating: 4.5 (1 reviews)
[EL003] Sony WH-1000XM5 Headphones	\$1845.00 8 pcs Available Rating: 4.9 (4 reviews)

=====

Number of products: 5

=====

=====

CATEGORY: BOOKS

=====

[BK001] Clean Code - Robert Martin	\$81.90 40 pcs Available Rating: 5.0 (3 reviews)
[BK003] The Witcher - Andrzej Sapkowski	\$44.10 50 pcs Available Rating: 4.9 (1 reviews)
[BK002] Sapiens - Yuval Harari	\$56.70 35 pcs Available Rating: 4.9 (2 reviews)
[BK004] Atomic Habits - James Clear	\$69.30 25 pcs Available Rating: 4.8 (2 reviews)

=====

Number of products: 4

=====

=====

ALL PRODUCTS

Sorting: by rating (best first)

=====

[BK001] Clean Code - Robert Martin	\$81.90 40 pcs Available Rating: 5.0 (3 reviews)
[EL005] Apple iPad Pro 12.9" Tablet	\$5488.88 -15% 2 pcs Low Stock Rating: 5.0 (2 reviews)
[EL003] Sony WH-1000XM5 Headphones	\$1845.00 8 pcs Available Rating: 4.9 (4 reviews)
[SP001] Trek Mountain Bike	\$4151.25 4 pcs Low Stock Rating: 4.9 (1 reviews)
[BK003] The Witcher - Andrzej Sapkowski	\$44.10 50 pcs Available Rating: 4.9 (1 reviews)
[EL001] Dell XPS 15 Laptop	\$5381.25 5 pcs Low Stock Rating: 4.9 (3 reviews)
[BK002] Sapiens - Yuval Harari	\$56.70 35 pcs Available Rating: 4.9 (2 reviews)
[CL004] Adidas Ultraboost Running Shoes	\$774.90 4 pcs Low Stock Rating: 4.9 (2 reviews)
[BK004] Atomic Habits - James Clear	\$69.30 25 pcs Available Rating: 4.8 (2 reviews)
[SP004] Electric Treadmill	\$2092.23 -30% 2 pcs Low Stock Rating: 4.7 (1 reviews)
[CL002] Levi's 501 Jeans	\$430.50 25 pcs Available Rating: 4.7 (1 reviews)
[EL002] Samsung Galaxy S23 Smartphone	\$3874.50 -10% 12 pcs Available Rating: 4.7 (2 reviews)
[HL002] Whey Protein WPC 1kg	\$116.64 -10% 30 pcs Available Rating: 4.6 (1 reviews)
[HM003] Wool Rug 200x300cm	\$1279.20 5 pcs Low Stock Rating: 4.6 (1 reviews)
[SP002] Manduka Yoga Mat	\$365.31 18 pcs Available Rating: 4.5 (1 reviews)
[EL004] LG UltraWide 34" Monitor	\$2767.50 3 pcs Low Stock Rating: 4.5 (1 reviews)
[HL001] Vitamin D3 4000 IU	\$40.50 60 pcs Available Rating: 4.5 (1 reviews)
[CL001] The North Face Winter Jacket	\$688.80 15 pcs Available Rating: 4.4 (2 reviews)

[HM004]	3-Seater Corner Sofa	\$2814.24 -20% 3 pcs Low Stock Rating: 4.4 (1 reviews)
[CL003]	Nike Sportswear Hoodie	\$294.46 -5% 30 pcs Available Rating: 4.4 (1 reviews)
[SP003]	Adjustable Dumbbells 2x20kg	\$581.18 12 pcs Available Rating: 4.3 (1 reviews)
[HM001]	Ergonomic Office Chair	\$959.40 10 pcs Available Rating: 4.2 (1 reviews)
[HL003]	Anti-Wrinkle Cream	\$194.40 15 pcs Available Rating: 4.2 (1 reviews)
[HM002]	LED Floor Lamp	\$239.85 20 pcs Available Rating: 4.0 (1 reviews)

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Total products: 24

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ALL PRODUCTS

Sorting: multi-criteria: category -> rating -> price -> name

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[EL005]	Apple iPad Pro 12.9" Tablet	\$5488.88 -15% 2 pcs Low Stock Rating: 5.0 (2 reviews)
[EL003]	Sony WH-1000XM5 Headphones	\$1845.00 8 pcs Available Rating: 4.9 (4 reviews)
[EL001]	Dell XPS 15 Laptop	\$5381.25 5 pcs Low Stock Rating: 4.9 (3 reviews)
[EL002]	Samsung Galaxy S23 Smartphone	\$3874.50 -10% 12 pcs Available Rating: 4.7 (2 reviews)
[EL004]	LG UltraWide 34" Monitor	\$2767.50 3 pcs Low Stock Rating: 4.5 (1 reviews)
[CL004]	Adidas Ultraboost Running Shoes	\$774.90 4 pcs Low Stock Rating: 4.9 (2 reviews)
[CL002]	Levi's 501 Jeans	\$430.50 25 pcs Available Rating: 4.7 (1 reviews)
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[CL001]	The North Face Winter Jacket	\$688.80 15 pcs Available Rating: 4.4 (2 reviews)
[HM003]	Wool Rug 200x300cm	\$1279.20 5 pcs Low Stock Rating: 4.6 (1 reviews)
[HM004]	3-Seater Corner Sofa	\$2814.24 -20% 3 pcs Low Stock Rating: 4.4 (1 reviews)
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[SP002]	Manduka Yoga Mat	\$365.31 18 pcs Available Rating: 4.5 (1 reviews)
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[BK001]	Clean Code - Robert Martin	\$81.90 40 pcs Available Rating: 5.0 (3 reviews)
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[BK002]	Sapiens - Yuval Harari	\$56.70 35 pcs Available Rating: 4.9 (2 reviews)
[BK004]	Atomic Habits - James Clear	\$69.30 25 pcs Available Rating: 4.8 (2 reviews)
[HL002]	Whey Protein WPC 1kg	\$116.64 -10% 30 pcs Available Rating: 4.6 (1 reviews)
[HL001]	Vitamin D3 4000 IU	\$40.50 60 pcs Available Rating: 4.5 (1 reviews)
[HL003]	Anti-Wrinkle Cream	\$194.40 15 pcs Available Rating: 4.2 (1 reviews)

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Total products: 24

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ALL PRODUCTS

Sorting: by potential profit (highest first)

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[EL002]	Samsung Galaxy S23 Smartphone	\$3874.50 -10% 12 pcs Available Rating: 4.7 (2 reviews)
[EL001]	Dell XPS 15 Laptop	\$5381.25 5 pcs Low Stock Rating: 4.9 (3 reviews)
[SP001]	Trek Mountain Bike	\$4151.25 4 pcs Low Stock Rating: 4.9 (1 reviews)
[EL003]	Sony WH-1000XM5 Headphones	\$1845.00 8 pcs Available Rating: 4.9 (4 reviews)

[CL002]	Levi's 501 Jeans	\$430.50 25 pcs Available Rating: 4.7 (1 reviews)
[CL001]	The North Face Winter Jacket	\$688.80 15 pcs Available Rating: 4.4 (2 reviews)
[HM001]	Ergonomic Office Chair	\$959.40 10 pcs Available Rating: 4.2 (1 reviews)
[CL003]	Nike Sportswear Hoodie	\$294.46 -5% 30 pcs Available Rating: 4.4 (1 reviews)
[EL004]	LG UltraWide 34" Monitor	\$2767.50 3 pcs Low Stock Rating: 4.5 (1 reviews)
[SP003]	Adjustable Dumbbells 2x20kg	\$581.18 12 pcs Available Rating: 4.3 (1 reviews)
[SP002]	Manduka Yoga Mat	\$365.31 18 pcs Available Rating: 4.5 (1 reviews)
[EL005]	Apple iPad Pro 12.9" Tablet	\$5488.88 -15% 2 pcs Low Stock Rating: 5.0 (2 reviews)
[HM003]	Wool Rug 200x300cm	\$1279.20 5 pcs Low Stock Rating: 4.6 (1 reviews)
[HM004]	3-Seater Corner Sofa	\$2814.24 -20% 3 pcs Low Stock Rating: 4.4 (1 reviews)
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[HL001]	Vitamin D3 4000 IU	\$40.50 60 pcs Available Rating: 4.5 (1 reviews)
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[BK003]	The Witcher - Andrzej Sapkowski	\$44.10 50 pcs Available Rating: 4.9 (1 reviews)
[BK002]	Sapiens - Yuval Harari	\$56.70 35 pcs Available Rating: 4.9 (2 reviews)
[BK004]	Atomic Habits - James Clear	\$69.30 25 pcs Available Rating: 4.8 (2 reviews)

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Total products: 24

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SELECTED PRODUCT DETAILS

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PRODUCT DETAILS - Electronics

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Code:	EL001
Name:	Dell XPS 15 Laptop
Manufacturer:	Dell
Category:	Electronics

Purchase price:	\$3500.00
Selling price:	\$5381.25
VAT (23%):	\$1006.25

Stock quantity:	5 pcs
Status:	Low Stock
Stock value:	\$17500.00
Potential profit:	\$9406.25

Delivery date:	04/08/2026
Days in stock:	0
Warranty:	24 months

Rating:	4.9 / 5.0
Review count:	3

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PRODUCT DETAILS - Electronics

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Code: EL002
Name: Samsung Galaxy S23 Smartphone
Manufacturer: Samsung
Category: Electronics

Purchase price: \$2800.00
Selling price: \$4305.00
Discount: -10%
Price after discount: \$3874.50
VAT (23%): \$724.50

Stock quantity: 12 pcs
Status: Available
Stock value: \$33600.00
Potential profit: \$12894.00

Delivery date: 04/08/2026
Days in stock: 0
Warranty: 24 months

Rating: 4.7 / 5.0
Review count: 2

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CREATING ORDERS

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Created: Order #0001 | John Doe | 04/08/2026 | \$9316.95 | Items: 3 | Status: Confirmed
Created: Order #0002 | Jane Smith | 04/08/2026 | \$4520.12 | Items: 3 | Status: Processing
Created: Order #0003 | Peter Johnson | 04/08/2026 | \$4516.56 | Items: 2 | Status: Shipped

=====

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ORDERS

=====

Order #0001 | John Doe | 04/08/2026 | \$9316.95 | Items: 3 | Status: Confirmed
Order #0002 | Jane Smith | 04/08/2026 | \$4520.12 | Items: 3 | Status: Processing
Order #0003 | Peter Johnson | 04/08/2026 | \$4516.56 | Items: 2 | Status: Shipped

=====

=====

SELECTED ORDER DETAILS

=====

=====

ORDER #0001

```
=====
Customer:           John Doe
Order date:         04/08/2026
Status:             Confirmed
=====
```

```
-----
Dell XPS 15 Laptop                                1 pcs
    $5381.25 x 1 = $5381.25
Sony WH-1000XM5 Headphones                        2 pcs
    $1845.00 x 2 = $3690.00
Clean Code - Robert Martin                        3 pcs
    $81.90 x 3 = $245.70
-----
```

```
TOTAL VALUE:           $9316.95
=====
```

```
=====
ANALYSIS USING ENUMSET
=====
```

```
Categories with margin > 30%:
```

- Clothing (40%)
- Sports & Recreation (35%)
- Health & Beauty (50%)

```
Categories requiring warranty:
```

- Electronics (24 months)
- Home & Garden (12 months)

```
Categories with reduced VAT rate (<20%):
```

- Books (5%)
 - Health & Beauty (8%)
- ```
=====
```

```
=====
PRODUCT SEARCH
=====
```

```
Sony products:
```

- [EL003] Sony WH-1000XM5 Headphones | \$1845.00 | 6 pcs | Available | Rating: 4.9 (4 reviews)

```
Product with code EL002:
```

- [EL002] Samsung Galaxy S23 Smartphone | \$3874.50 -10% | 11 pcs | Available | Rating: 4.7 (2 reviews)
- ```
=====
```

```
ENUMMAP DEMONSTRATION
=====
```

```
Number of products in each category:
```

- Electronics : 5 products
- Clothing : 4 products

```

Home & Garden      : 4 products
Sports & Recreation : 4 products
Books              : 4 products
Health & Beauty    : 3 products

```

DEMONSTRATION OF equals(), hashCode() and Comparable

```

p1 (EL001): Dell XPS 15 Laptop
p2 (EL002): Samsung Galaxy S23 Smartphone
p3 (EL001): Dell XPS 15 Laptop

```

```

p1 == p3:          true
p1.equals(p3):     true
p1.hashCode():     66034761
p3.hashCode():     66034761

```

```

p1.compareTo(p2):  -15 (alphabetic sorting: Dell XPS 15 Laptop vs Samsung Galaxy S23 Smartphone)

```

ENUM METHODS DEMONSTRATION

```

All categories (values()):
- ELECTRONICS -> Electronics
- CLOTHING -> Clothing
- HOME -> Home & Garden
- SPORTS -> Sports & Recreation
- BOOKS -> Books
- HEALTH -> Health & Beauty

```

```

Ordinal (position in enum):
Electronics: ordinal = 0
Clothing: ordinal = 1
Home & Garden: ordinal = 2
Sports & Recreation: ordinal = 3
Books: ordinal = 4
Health & Beauty: ordinal = 5

```

```

valueOf() - String to Enum conversion:
Category.valueOf("ELECTRONICS") = Electronics
Category.findByName("Books") = Books

```

ENUM AvailabilityStatus DEMONSTRATION

```

Quantity  0 pcs -> Out of Stock
Quantity  2 pcs -> Low Stock
Quantity  5 pcs -> Low Stock
Quantity 10 pcs -> Available

```

Quantity 100 pcs -> Available

```
=====
ENUM DiscountType DEMONSTRATION
=====
```

Base price: \$1000

No discount	-> \$1000.00
-5%	-> \$950.00
-10%	-> \$900.00
-15%	-> \$850.00
-20%	-> \$800.00
-30%	-> \$700.00
-50%	-> \$500.00
-70%	-> \$300.00

19. ★ Milestone 2: present the result of your work to the instructor